

1. Limits

Throughout these notes, a function whose limit at x_0 is considered is assumed to be defined on an open interval containing x_0 , except possibly at x_0 itself. Its domain is denoted by D .

Definition. We say that $\lim_{x \rightarrow x_0} f(x) = L$ if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that, for every $x \in D$,

$$0 < |x - x_0| < \delta \implies |f(x) - L| < \varepsilon.$$

Thus a prescribed tolerance ε for the function values can be ensured by restricting x to a sufficiently small distance δ from x_0 , with $x \neq x_0$.

The order of the choices is essential:

$$\text{for every } \varepsilon > 0, \quad \text{there exists } \delta > 0, \quad \text{for every } x \in D.$$

The number δ may depend on ε , the function and the fixed numbers x_0, L , but not on the subsequently chosen x . A smaller positive δ also works.

The condition $0 < |x - x_0|$ excludes $x = x_0$. Consequently, the value $f(x_0)$ is not used in this definition; it may even be undefined.

Example

For the linear function $f(x) = 3x - 2$, we have

$$\lim_{x \rightarrow 1} (3x - 2) = 1.$$

To prove this, let $\varepsilon > 0$ and choose $\delta = \varepsilon/3$. If $0 < |x - 1| < \delta$, then

$$|(3x - 2) - 1| = 3|x - 1| < 3\delta = \varepsilon.$$

This is exactly the required estimate.

Example

For a constant function $f(x) = c$,

$$\lim_{x \rightarrow x_0} c = c.$$

Given any $\varepsilon > 0$, choose any $\delta > 0$, for instance $\delta = 1$. For every x in the required neighborhood,

$$|f(x) - c| = |c - c| = 0 < \varepsilon.$$

Example

We prove

$$\lim_{x \rightarrow 2} x^2 = 4.$$

Factoring the error gives

$$|x^2 - 4| = |x - 2| |x + 2|.$$

To control the second factor, first require $|x - 2| < 1$. Then

$$|x + 2| = |(x - 2) + 4| \leq |x - 2| + 4 < 5.$$

Given $\varepsilon > 0$, choose

$$\delta = \min\{1, \varepsilon/5\}.$$

If $0 < |x - 2| < \delta$, both restrictions hold, and

$$|x^2 - 4| < 5|x - 2| < 5\delta \leq \varepsilon.$$

This proves the limit. The restriction $\delta \leq 1$ gives a fixed bound on $|x + 2|$; the restriction $\delta \leq \varepsilon/5$ makes the final error small enough. Choosing δ as the minimum ensures that both requirements hold.

2. Uniqueness of limits

Theorem. If

$$\lim_{x \rightarrow x_0} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow x_0} f(x) = M,$$

then $L = M$.

Proof. Suppose instead that $L \neq M$, and set $\varepsilon = |L - M|/2 > 0$. The two limit statements give positive numbers δ_L, δ_M such that

$$0 < |x - x_0| < \delta_L \implies |f(x) - L| < \varepsilon,$$

$$0 < |x - x_0| < \delta_M \implies |f(x) - M| < \varepsilon.$$

Choose $r > 0$ so that f is defined for $0 < |x - x_0| < r$, and put $\delta = \min\{r, \delta_L, \delta_M\}$. There is a point in this punctured interval, for example $x = x_0 + \delta/2$. Both estimates hold there. By the triangle inequality,

$$|L - M| \leq |L - f(x)| + |f(x) - M| < 2\varepsilon = |L - M|,$$

which is impossible. Therefore $L = M$.

3. Limits of sums

Theorem. Suppose f and g are defined near x_0 , except possibly at x_0 , and

$$\lim_{x \rightarrow x_0} f(x) = L, \quad \lim_{x \rightarrow x_0} g(x) = M.$$

Then $\lim_{x \rightarrow x_0} (f(x) + g(x)) = L + M$.

Proof. Let $\varepsilon > 0$. The assumed limits give $\delta_f, \delta_g > 0$ such that

$$\begin{aligned} 0 < |x - x_0| < \delta_f &\implies |f(x) - L| < \varepsilon/2, \\ 0 < |x - x_0| < \delta_g &\implies |g(x) - M| < \varepsilon/2. \end{aligned}$$

Choose $\delta = \min\{\delta_f, \delta_g\}$. If $0 < |x - x_0| < \delta$ and x is in the common domain, then

$$\begin{aligned} |(f(x) + g(x)) - (L + M)| &\leq |f(x) - L| + |g(x) - M| \\ &< \varepsilon/2 + \varepsilon/2 = \varepsilon. \end{aligned}$$

This proves the sum rule. The minimum makes both estimates available at the same x ; the two tolerances $\varepsilon/2$ keep their sum below the prescribed ε .

4. Negating the limit statement

For a fixed L , the statement $\lim_{x \rightarrow x_0} f(x) = L$ is false exactly when there exists $\varepsilon_0 > 0$ such that, for every $\delta > 0$, there exists $x \in D$ satisfying

$$\boxed{0 < |x - x_0| < \delta \quad \text{and} \quad |f(x) - L| \geq \varepsilon_0.}$$

In quantifier notation, the negation is

$$\exists \varepsilon_0 > 0 \quad \forall \delta > 0 \quad \exists x \in D : \quad 0 < |x - x_0| < \delta \quad \text{and} \quad |f(x) - L| \geq \varepsilon_0.$$

There is a fixed positive tolerance that cannot be ensured, however small a neighborhood we choose. The point x may depend on δ , but ε_0 is chosen first and must work for every δ .

Negating an implication produces its hypothesis together with the negation of its conclusion. This is why the condition on x remains $0 < |x - x_0| < \delta$, whereas $|f(x) - L| < \varepsilon$ becomes $|f(x) - L| \geq \varepsilon_0$.

Showing that a particular L is not the limit does not show that no limit exists. To prove that there is no finite limit, one must rule out every $L \in \mathbb{R}$. The tolerance used to rule out a candidate may depend on that candidate.

Reading. Stewart, *Calculus: Early Transcendentals*, 9th ed., §2.4 for the precise definition and §2.3 for the sum law.